Khaoula SOUMANE

Data Engineer | Data Analyst

PROFILE

Recent Data Science and Data Engineering graduate with skills in data engineering, analytics, and machine learning. Seeking a full-time position to contribute to data-driven projects and further develop my technical expertise.

CONTACT

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Khaoula Soumane

EDUCATION

Master's Degree in High Performance Computing & Big Data
ENSEEIHT
2025 – 2026

Engineering Degree in Data Science & Data Engineering
INPT
2023 – 2026

TECHNOLOGIES

Data Engineering:
ETL / ELT, Data Pipelines, Data Quality & Observability, FinOps

Cloud & Big Data:
GCP, AWS, Apache Kafka, Apache Spark, Hadoop, MapReduce

Languages & Frameworks:
Python, SQL (PostgreSQL), Java, Scala, Django, FastAPI, Linux

MLOps:
Docker, Terraform, CI/CD, Git, Dynatrace

Data Science & Visualization:
Power BI, Tableau, Scikit-learn, TensorFlow, Matplotlib, Seaborn

Collaboration:
Agile (Scrum), Jira, Confluence, Documentation, MS Office

LANGUAGES

French: Fluent
English: C2 (TOEIC)
Arabic: Native

EXPERIENCE

AIR FRANCE - Data Engineer — Mar – Sep 2026

- **Cloud ELT Pipelines:** Automated the ingestion and transformation of large-scale data streams (*8 TB/day*) using BigQuery, Kafka, Cloud Storage, and Workflows.
- **Observability & FinOps:** Benchmarked monitoring tools and centralized GCP metrics using Dynatrace and Power BI to monitor data processing jobs and analyze cloud costs.
- **Anomaly Detection & Root Cause Analysis:** Developed statistical and machine learning approaches to detect anomalies in BigQuery and Dataform costs and support the identification of pipeline incidents.
- **Communication:** Produced technical documentation, managed code using Git, and presented findings to stakeholders.

IRIT – CRCT TOULOUSE - Data Science & Software Engineering Project — Nov 2025 – Feb 2026

- **Software Architecture & Scientific Collaboration:** Conducted a literature review in collaboration with the Cancer Research Center of Toulouse and designed a modular platform using FastAPI and React to integrate ML/DL models.
- **Multimodal Data Processing:** Developed MRI and MRSI spectroscopy processing pipelines, including 3D spatial filtering and spectral processing.
- **Visualization & Multimodal Fusion:** Developed interactive tools for exploring brain images and aligning anatomical and spectroscopic data.
- **Metabolite Quantification:** Automated metabolite concentration calculations, generated heatmaps, and exported structured results.

NSAYBLIK - MLOps Intern — Jul – Aug 2025

- **End-to-End Data/ML Pipeline:** Automated data ingestion, OCR extraction using PyTesseract, data preparation, and profile classification from PDF resumes and images.
- **ML Deployment & Operationalization:** Deployed the model through a Django web application, containerized with Docker and served using Nginx and Gunicorn, with automated CI/CD workflows.

PERSONAL PROJECTS

Multi-Agent Intelligent Tutoring System – AI Engineering
Designed an LLM- and RAG-based multi-agent system using LangChain to generate personalized educational content and incorporate user feedback.

Distributed Processing with Hadoop & MapReduce – Data Engineering
Deployed a Hadoop cluster (HDFS, YARN) and developed Java MapReduce jobs for large-scale distributed data processing.

LEADERSHIP & ENGAGEMENT

Debate Club: Organized workshops and coordinated team activities.
Safer Internet Day: Volunteered to raise awareness of the responsible use of AI.

RELEVANT COURSEWORK

Statistics & Mathematics:
Statistics, Probability, Stochastic Processes, Bayesian Methods.
Data Science & AI:
Machine Learning, Deep Learning, Data Mining, Signal Processing.
Data Engineering & Systems:
Big Data Processing, Cloud Computing, Parallel Computing, Distributed Systems.
Geographic Information Systems:
QGIS.